



Started on Monday, 17 August 2026, 7:55 PM

State Finished

Completed on Monday, 17 August 2026, 9:35 PM

Time taken 1 hour 40 mins

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int n, rev = 0, remainder;
5      int count = 0;
6      scanf("%d", &n);
7
8      rev = 0;
9      count++;
10
11     while (1) {
12         count++;
13         if (n == 0) break;
14
15         remainder = n % 10;
16         count++;
17
18         rev = rev * 10 + remainder;
19         count++;
20
21         n /= 10;
22         count++;
23     }
24     count++;
25     printf("%d\n", count);
26     return 0;
27 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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